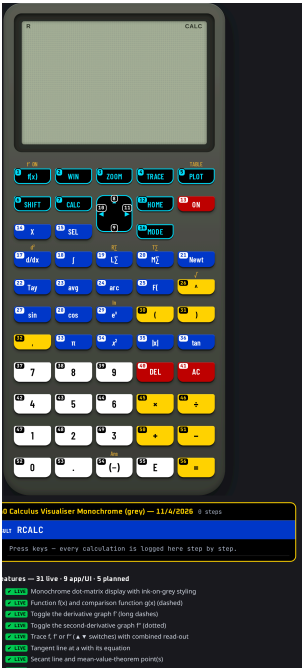
U-45 CALCULATORS

User Manual

Calculator #40 Calculus Visualiser Monochrome (grey)

Graphing – monochrome



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#40 — Calculus Visualiser Monochrome (grey)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example **#1 f(x)**. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

calculus visualiser on a monochrome dot-matrix: derivatives, Riemann sums, Newton, Taylor.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers

12. Appendix B — Function finder (A–Z)
13. Appendix C — Glossary
14. Appendix D — Model notes & limits
15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#40** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#37, #38, #39, #42, #43, #44, #47, #48, #49, #52, #53, #54, #55
op	Operators & execute	#26, #30, #31, #32, #45, #46, #50, #51, #56
fn	Functions	#14, #15, #17, #18, #19, #20, #21, #22, #23, #24, #25, #27, #28, #29, #33, #34, #35, #36
mem	Memory, variables & registers	—
clr	Clear, delete & power	#13, #40, #41

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #7, #8, #9, #10, #11, #12, #16**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE Monochrome dot-matrix display with ink-on-grey styling
2. LIVE Function f(x) and comparison function g(x) (dashed)
3. LIVE Toggle the derivative graph f′ (long dashes)
4. LIVE Toggle the second-derivative graph f″ (dotted)
5. LIVE Trace f, f′ or f″ (▲ ▼ switches) with combined read-out
6. LIVE Tangent line at a with its equation
7. LIVE Secant line and mean-value-theorem point(s)
8. LIVE Left Riemann sum with drawn rectangles
9. LIVE Right Riemann sum
10. LIVE Midpoint Riemann sum
11. LIVE Trapezoid rule with drawn trapezoids
12. LIVE Riemann error against the exact integral
13. LIVE Definite integral with shaded region
14. LIVE Area between f and g (absolute)
15. LIVE Newton's method with drawn tangent steps
16. LIVE Taylor polynomial up to degree 6 overlaid
17. LIVE Taylor coefficients listed
18. LIVE Roots, extrema and inflection points in the window
19. LIVE Average value of f on [a,b]
20. LIVE Arc length of y = f(x)

21. LIVE Clear all overlays
22. LIVE Table of x, f, f′ , f″
23. LIVE Home F(x), der(x), der2(x)
24. LIVE Home integ(a,b) by Simpson's rule (1000 panels)
25. LIVE Home lsum(), rsum(), msum(), tsum()
26. LIVE Home newt(x_0) root finder
27. LIVE Home taylor(a,n,x) value
28. LIVE Home avgv(a,b) and arclen(a,b)
29. LIVE f′ quick-toggle key
30. LIVE Zoom menu with square/integer/decimal
31. LIVE Trace jump to x with EXE
32. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
33. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle
34. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
35. APP/UI Show / hide key numbers with one button (remembered between sessions)
36. APP/UI Copy the current display text to the clipboard
37. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
38. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
39. APP/UI Reference photo viewer and tool-tip name on every key

40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Graph image export (PNG)**
42. PLANNED **Shareable function list link**
43. PLANNED **Dark screen theme**
44. PLANNED **Speech read-out of values**
45. PLANNED **Exam mode**

2b · Feature guide

Calculator #40 is a **calculus visualiser on a monochrome dot-matrix: derivatives, Riemann sums, Newton, Taylor**. Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Monochrome dot-matrix display with ink-on-grey styling

Where: #16 MODE SETUP — angle unit, number format, display options

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Function f(x) and comparison function g(x) (dashed)

Where: #6 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function; #32 , Comma / argument separator (between function arguments, list items or coordinates)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Toggle the derivative graph f′ (long dashes)

Where: #1 f(x) Y= — enter functions to graph; #2 WIN WINDOW — set the graph view; #4 TRACE TRACE — move along a graph

See it in action: Derivative of $x^2 - 2$ at x = 1.5 — *der(1.5)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Toggle the second-derivative graph f″ (dotted)

Where: #1 f(x) Y= — enter functions to graph; #17 d/dx Derivative of f; #2 WIN WINDOW — set the graph view

See it in action: Derivative of $x^2 - 2$ at x = 1.5 — *der(1.5)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Trace f, f′ or f″ (▲ ▼ switches) with combined read-out

Where: #4 TRACE **TRACE — move along a graph**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Tangent line at a with its equation

Where: #7 CALC **Riemann, Newton, Taylor, tangent;** #8 ▲ **Cursor ▲ — up / previous history line / previous item;** #9 ▼ **Cursor ▼ — down / next history line / next item**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Secant line and mean-value-theorem point(s)

Where: #8 ▲ Cursor ▲ — up / previous history line / previous item; #9 ▼ Cursor ▼ — down / next history line / next item; #10 ◀ Cursor ◀ — move left in the line

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Left Riemann sum with drawn rectangles

Where: #19 L∑ Left Riemann sum; #7 CALC Riemann, Newton, Taylor, tangent; #10 ◀ Cursor ◀ — move left in the line

See it in action: Left Riemann sum, 4 strips on [0,2] — *lsum(0,2,4)*= -2.25; Midpoint sum, 4 strips on [0,2] — *msum(0,2,4)*= -1.375

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Right Riemann sum

Where: #19 L∑ Left Riemann sum; #7 CALC Riemann, Newton, Taylor, tangent; #11 ▶ Cursor ▶ — move right in the line

See it in action: Left Riemann sum, 4 strips on [0,2] — *lsum(0,2,4)*= -2.25; Midpoint sum, 4 strips on [0,2] — *msum(0,2,4)*= -1.375

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Midpoint Riemann sum

Where: #19 L∑ Left Riemann sum; #20 M∑ Midpoint sum; #7 CALC Riemann, Newton, Taylor, tangent

See it in action: Left Riemann sum, 4 strips on [0,2] — *lsum(0,2,4)*= -2.25; Midpoint sum, 4 strips on [0,2] — *msum(0,2,4)*= -1.375

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Trapezoid rule with drawn trapezoids

Where: #20 M∑ Midpoint sum

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Riemann error against the exact integral

Where: #7 CALC Riemann, Newton, Taylor, tangent; #18 ∫ Definite integral of f; #19 L∑ Left Riemann sum

See it in action: Left Riemann sum, 4 strips on [0,2] — *lsum(0,2,4)* = -2.25

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Definite integral with shaded region

Where: #18 ∫ Definite integral of f

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Area between f and g (absolute)

Where: #32 , Comma / argument separator (between function arguments, list items or coordinates); #35 |x| Absolute value |x|

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Newton's method with drawn tangent steps

Where: #7 CALC Riemann, Newton, Taylor, tangent; #21 Newt Newton root from x_0; #36 tan Tangent (current angle unit)

See it in action: Newton's method for $x^2 - 2$ from 1 ($\sqrt{2}$) — *newt(1)*= 1.41421356

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Taylor polynomial up to degree 6 overlaid

Where: #22 Tay Taylor polynomial value; #7 CALC Riemann, Newton, Taylor, tangent

See it in action: Degree-2 Taylor polynomial about 1, evaluated at 3 — *taylor(1,2,3)* = 7

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Taylor coefficients listed

Where: #7 CALC Riemann, Newton, Taylor, tangent; #22 Tay Taylor polynomial value

See it in action: Degree-2 Taylor polynomial about 1, evaluated at 3 — *taylor(1,2,3)* = 7

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Roots, extrema and inflection points in the window

Where: #2 WIN **WINDOW — set the graph view**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Average value of f on [a,b]

Where: #23 avg **Average value**

See it in action: Average value of $x^2 - 2$ on [0,3] — *avgv(0,3)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Arc length of y = f(x)

Where: #24 arc **Arc length**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Clear all overlays

Where: #41 AC **AC — clear all of the current calculation**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Table of x, f, f′ , f″

Where: #5 PLOT **GRAPH — draw the graph**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Home F(x), der(x), der2(x)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Derivative of $x^2 - 2$ at x = 1.5 — *der(1.5)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Home integ(a,b) by Simpson's rule (1000 panels)

Where: #20 M∑ Midpoint sum

See it in action: $\int_0^3 (x^2 - 2)\,dx$ — *integ(0,3)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Home Isum(), rsum(), msum(), tsum()

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Left Riemann sum, 4 strips on [0,2] — *Isum(0,2,4)*= -2.25; Midpoint sum, 4 strips on [0,2] — *msum(0,2,4)*= -1.375

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Home newt(x_0) root finder

Where: #21 Newt Newton root from x_0; #26 ^ Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

See it in action: Newton's method for $x^2 - 2$ from 1 ($\sqrt{2}$) — *newt(1)*= 1.41421356

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Home taylor(a,n,x) value

Where: #7 CALC Riemann, Newton, Taylor, tangent; #22 Tay Taylor polynomial value

See it in action: Degree-2 Taylor polynomial about 1, evaluated at 3 — *taylor(1,2,3)* = 7

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Home avgv(a,b) and arclen(a,b)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Average value of $x^2 - 2$ on [0,3] — *avgv(0,3)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 f′ quick-toggle key

Where: #1 f(x) Y= — enter functions to graph

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Zoom menu with square/integer/decimal

Where: #3 ZOOM ZOOM — zoom presets; #26 ^ Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512); #34 x^2 Square x^2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Trace jump to x with EXE

Where: #4 TRACE TRACE — move along a graph; #56 = Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Try it: Press Reset, use the key(s) above, and write the display here:

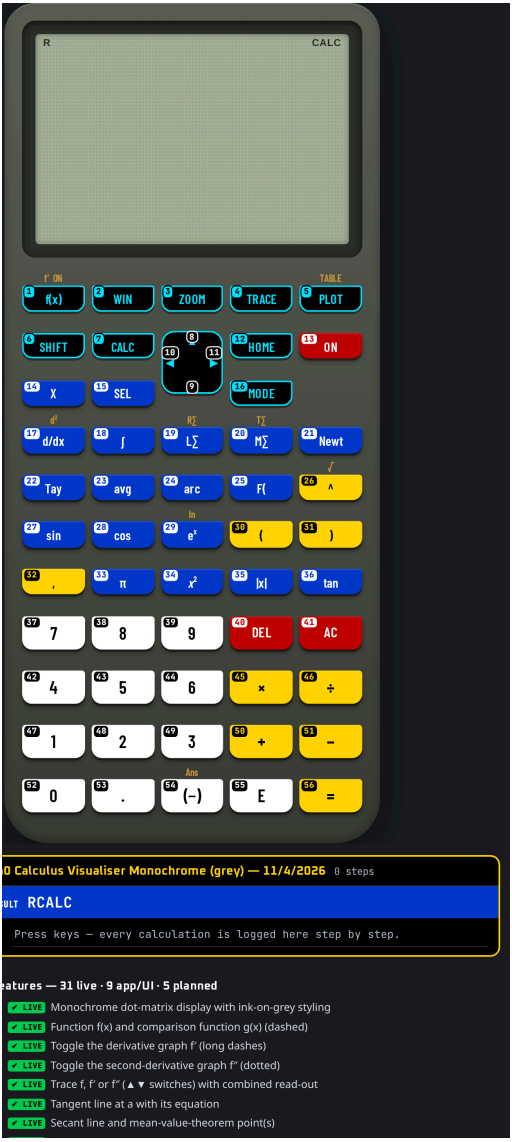
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #40 with key numbers shown (Hi-Contrast colours).

Key table

"Shift" = 2nd / f / SHIFT layer; "Alpha" = ALPHA layer; "3rd" = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	f(x)	f' ON	·	·	mod
2	WIN	·	·	·	mod
3	ZOOM	·	·	·	mod
4	TRACE	·	·	·	mod
5	PLOT	TABLE	·	·	mod
6	SHIFT	·	·	·	mod
7	CALC	·	·	·	mod
8	▲	·	·	·	mod
9	▼	·	·	·	mod
10	◀	·	·	·	mod
11	▶	·	·	·	mod
12	HOME	·	·	·	mod
13	ON	·	·	·	clr
14	X	·	·	·	fn
15	SEL	·	·	·	fn
16	MODE	·	·	·	mod
17	d/dx	d^2	·	·	fn
18	∫	·	·	·	fn
19	L∑	R∑	·	·	fn
20	M∑	T∑	·	·	fn
21	Newt	·	·	·	fn
22	Tay	·	·	·	fn
23	avg	·	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	arc	·	·	·	fn
25	F(	·	·	·	fn
26	^	√	·	·	op
27	sin	·	·	·	fn
28	cos	·	·	·	fn
29	ex	ln	·	·	fn
30	(	·	·	·	op
31	)	·	·	·	op
32	,	·	·	·	op
33	π	·	·	·	fn
34	x^2	·	·	·	fn
35	|x|	·	·	·	fn
36	tan	·	·	·	fn
37	7	·	·	·	num
38	8	·	·	·	num
39	9	·	·	·	num
40	DEL	·	·	·	clr
41	AC	·	·	·	clr
42	4	·	·	·	num
43	5	·	·	·	num
44	6	·	·	·	num
45	×	·	·	·	op
46	÷	·	·	·	op
47	1	·	·	·	num
48	2	·	·	·	num
49	3	·	·	·	num
50	+	·	·	·	op
51	−	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
52	0	·	·	·	num
53	.	·	·	·	num
54	(−)	Ans	·	·	num
55	E	·	·	·	num
56	=	·	·	·	op

4 · Key-by-key reference

Every key on calculator #40, in number order, with every function it carries in this app.

#1 f(x) `Shift, modes, menus & navigation`

Primary: Y= — enter functions to graph

shifted (2nd / f / SHIFT layer) (press #6 first): f′ ON — Toggle derivative graph

#2 WIN `Shift, modes, menus & navigation`

Primary: WINDOW — set the graph view

#3 ZOOM `Shift, modes, menus & navigation`

Primary: ZOOM — zoom presets

#4 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

#5 PLOT `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #6 first): TABLE — TABLE — table of values

#6 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 CALC `Shift, modes, menus & navigation`

Primary: Riemann, Newton, Taylor, tangent

#8 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#9 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#10 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#11 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#12 HOME `Shift, modes, menus & navigation`

Primary: HOME — return to the main calculation screen

#13 ON `Clear, delete & power`

Primary: ON — power on

#14 X Functions

Primary: Variable x

#15 SEL Functions

Primary: sel

#16 MODE Shift, modes, menus & navigation

Primary: SETUP — angle unit, number format, display options

#17 d/dx Functions

Primary: Derivative of f

shifted (2nd / f / SHIFT layer) (press #6 first): d^2 — Second derivative

#18 ∫ Functions

Primary: Definite integral of f

#19 L∑ Functions

Primary: Left Riemann sum

shifted (2nd / f / SHIFT layer) (press #6 first): R∑ — Right Riemann sum

#20 M∑ Functions

Primary: Midpoint sum

shifted (2nd / f / SHIFT layer) (press #6 first): T∑ — Trapezoid rule

#21 Newt Functions

Primary: Newton root from x_0

#22 Tay Functions

Primary: Taylor polynomial value

#23 avg Functions

Primary: Average value

#24 arc Functions

Primary: Arc length

#25 F(Functions

Primary: Evaluate f

#26 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #6 first): √ — Square root √

Keyboard: [^]

#27 sin Functions

Primary: Sine (uses the current angle unit)

Keyboard: [s]

#28 cos Functions

Primary: Cosine (current angle unit)

Keyboard: [c]

#29 ex Functions

Primary: e^x — exponential

shifted (2nd / f / SHIFT layer) (press #6 first): ln — Natural logarithm ln

#30 (Operators & execute

Primary: Opening bracket (

Keyboard: [(]

#31) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#32 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#33 π Functions

Primary: π (3.14159265…) as an exact constant

Keyboard: p

#34 x^2 Functions

Primary: Square x^2

Keyboard: q

#35 |x| Functions

Primary: Absolute value |x|

#36 tan Functions

Primary: Tangent (current angle unit)

Keyboard: t

#37 7 `Digits & number entry`

Primary: Digit 7

Keyboard: `7`

#38 8 `Digits & number entry`

Primary: Digit 8

Keyboard: `8`

#39 9 `Digits & number entry`

Primary: Digit 9

Keyboard: `9`

#40 DEL `Clear, delete & power`

Primary: DEL — delete the character at the cursor

Keyboard: `Backspace` `Delete`

#41 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: `Escape`

#42 4 `Digits & number entry`

Primary: Digit 4

Keyboard: `4`

#43 **5** Digits & number entry

Primary: Digit 5

Keyboard: 5

#44 **6** Digits & number entry

Primary: Digit 6

Keyboard: 6

#45 **×** Operators & execute

Primary: Multiplication ×

Keyboard: * x

#46 **÷** Operators & execute

Primary: Division ÷

Keyboard: /

#47 **1** Digits & number entry

Primary: Digit 1

Keyboard: 1

#48 **2** Digits & number entry

Primary: Digit 2

Keyboard: 2

#49 **3** `Digits & number entry`

Primary: Digit 3

Keyboard: `3`

#50 **+** `Operators & execute`

Primary: Addition +

Keyboard: `+`

#51 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: `-`

#52 **0** `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#53 **.** `Digits & number entry`

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

Keyboard: `.`

#54 (−) Digits & number entry

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): Ans — Ans — recall the last answer into the current expression

#55 E Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#56 = Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: Enter =

5 · Worked examples

7 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Derivative of $x^2 - 2$ at x = 1.5

Calculate **der(1.5)**. Press **Reset** first.

#17 d/dx → #47 1 → #53 . → #43 5 → #31) → #56 =

Display shows 3 — expected 3. ✔ verified on the app.

Example 5.2 — $\int_0^3 (x^2 - 2)\,dx$

Calculate **integ(0,3)**. Press **Reset** first.

#18 ∫ → #52 0 → #32 , → #49 3 → #31) → #56 =

Display shows 3 — expected 3. ✔ verified on the app.

Example 5.3 — Left Riemann sum, 4 strips on [0,2]

Calculate **lsum(0,2,4)**. Press **Reset** first.

#19 L∑ → #52 0 → #32 , → #48 2 → #32 , → #42 4 → #31) → #56 =

Display shows -2.25 — expected -2.25. ✔ verified on the app.

Example 5.4 — Midpoint sum, 4 strips on [0,2]

Calculate **msum(0,2,4)**. Press **Reset** first.

#20 M∑ → #52 0 → #32 , → #48 2 → #32 , → #42 4 → #31) → #56 =

Display shows `-1.375` — expected -1.375. ✔ verified on the app.

Example 5.5 — Newton's method for $x^2 - 2$ from 1 ($\sqrt{2}$)

Calculate **newt(1)**. Press **Reset** first.

#21 Newt → #47 1 → #31) → #56 =

Display shows `1.414213562` — expected 1.41421356. ✔ verified on the app.

Example 5.6 — Degree-2 Taylor polynomial about 1, evaluated at 3

Calculate **taylor(1,2,3)**. Press **Reset** first.

#22 Tay → #47 1 → #32 , → #48 2 → #32 , → #49 3 → #31) → #56 =

Display shows `6.999999999` — expected 7. ✔ verified on the app.

Example 5.7 — Average value of $x^2 - 2$ on [0,3]

Calculate **avgv(0,3)**. Press **Reset** first.

#23 avg → **#52** 0 → **#32** , → **#49** 3 → **#31**) → **#56** =

Display shows `1` — expected 1. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

8 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #1 → f′ ON; #6 then #5 → TABLE; #6 then #17 → d^2; #6 then #19 → R∑.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #56.

Hack 18: Type faster on a computer

28 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
^	#26	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
s	#27	sin	Sine (uses the current angle unit)
c	#28	cos	Cosine (current angle unit)
(	#30	(	Opening bracket (
)	#31	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#32	,	Comma / argument separator (between function arguments, list items or coordinates)
p	#33	π	π (3.14159265…) as an exact constant
q	#34	x^2	Square x^2
t	#36	tan	Tangent (current angle unit)
7	#37	7	Digit 7
8	#38	8	Digit 8
9	#39	9	Digit 9
Backspace Delete	#40	DEL	DEL — delete the character at the cursor
Escape	#41	AC	AC — clear all of the current calculation
4	#42	4	Digit 4
5	#43	5	Digit 5
6	#44	6	Digit 6
* x	#45	×	Multiplication ×
/	#46	÷	Division ÷
1	#47	1	Digit 1

2	#48	2	Digit 2
3	#49	3	Digit 3
+	#50	+	Addition +
-	#51	–	Subtraction –
0	#52	0	Digit 0
.	#53	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#55	E	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#56	=	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (7)

Work each one on calculator #40, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **der(1.5)** = _____
2. **integ(0,3)** = _____
3. **lsum(0,2,4)** = _____
4. **msum(0,2,4)** = _____
5. **newt(1)** = _____
6. **taylor(1,2,3)** = _____
7. **avgv(0,3)** = _____

Set B — Key-finding drills (54)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. **Which key is "f(x)" (Y=)?** _____
2. **Which key is "WIN" (WINDOW)?** _____
3. **Which key is "ZOOM" (ZOOM)?** _____
4. **Which key is "TRACE" (TRACE)?** _____
5. **Which key is "PLOT" (GRAPH)?** _____
6. **Which key is "SHIFT" (SHIFT / 2nd)?** _____
7. **Which key is "CALC" (Riemann, Newton, Taylor, tangent)?** _____

8. Which key is “▲” (Cursor ▲)? _____

9. Which key is “▼” (Cursor ▼)? _____

10. Which key is “◀” (Cursor ◀)? _____

11. Which key is “▶” (Cursor ▶)? _____

12. Which key is “HOME” (HOME)? _____

13. Which key is “ON” (ON)? _____

14. Which key is “X” (Variable x)? _____

15. Which key is “SEL” (sel)? _____

16. Which key is “MODE” (SETUP)? _____

17. Which key is “d/dx” (Derivative of f)? _____

18. Which key is “∫” (Definite integral of f)? _____

19. Which key is “L∑” (Left Riemann sum)? _____

20. Which key is “M∑” (Midpoint sum)? _____

21. Which key is “Newt” (Newton root from x_0)? _____

22. Which key is “Tay” (Taylor polynomial value)? _____

23. Which key is “avg” (Average value)? _____

24. Which key is “arc” (Arc length)? _____

25. Which key is “F(” (Evaluate f)? _____

26. Which key is “^” (Power x^y / ^)? _____

27. Which key is “sin” (Sine)? _____

28. Which key is “cos” (Cosine)? _____

29. Which key is “ex” (e^x)? _____

30. Which key is “(” (Opening bracket)? _____

31. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

32. Which key is “,” (Comma / argument separator)? _____

33. Which key is “π” (π)? _____

34. Which key is “x^2” (Square x^2)? _____

35. Which key is “|x|” (Absolute value |x|)? _____

36. Which key is “tan” (Tangent)? _____

37. Which key is “DEL” (DEL)? _____

38. Which key is “AC” (AC)? _____

39. Which key is “×” (Multiplication ×)? _____

40. Which key is “÷” (Division ÷)? _____

41. Which key is “+” (Addition +)? _____

42. Which key is “−” (Subtraction −)? _____

43. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

44. Which key is “(−)” (Negative sign)? _____

45. Which key is “E” (×10^x / EXP / EE)? _____

46. Which key is “=” (Execute / =)? _____

47. How do you reach “f’ ON”? _____

48. How do you reach “TABLE”? _____

49. How do you reach “d^2”? _____

50. How do you reach “R∑”? _____

51. How do you reach “T∑”? _____

52. How do you reach “√”? _____

53. How do you reach “ln”? _____

54. How do you reach “Ans”? _____

Appendix A — Drill answers

Set A

1. **3** (display `3`)

   #17 d/dx → #47 1 → #53 . → #43 5 → #31) → #56 =

2. **3** (display `3`)

   #18 ∫ → #52 0 → #32 , → #49 3 → #31) → #56 =

3. **-2.25** (display `-2.25`)

   #19 L∑ → #52 0 → #32 , → #48 2 → #32 , → #42 4 → #31) → #56 =

4. **-1.375** (display `-1.375`)

   #20 M∑ → #52 0 → #32 , → #48 2 → #32 , → #42 4 → #31) → #56 =

5. **1.41421356** (display `1.414213562`)

   #21 Newt → #47 1 → #31) → #56 =

6. **7** (display `6.999999999`)

   #22 Tay → #47 1 → #32 , → #48 2 → #32 , → #49 3 → #31) → #56 =

7. **1** (display `1`)

   #23 avg → #52 0 → #32 , → #49 3 → #31) → #56 =

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #40
38. #41
39. #45
40. #46
41. #50
42. #51
43. #53
44. #54
45. #55
46. #56
47. #6 SHIFT then #1 f(x)
48. #6 SHIFT then #5 PLOT
49. #6 SHIFT then #17 d/dx
50. #6 SHIFT then #19 L∑
51. #6 SHIFT then #20 M∑
52. #6 SHIFT then #26 ^
53. #6 SHIFT then #29 ex
54. #6 SHIFT then #54 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	E	#55
Absolute value |x|	|x|	#35
AC — clear all of the current calculation	AC	#41
Addition +	+	#50
Ans — recall the last answer into the current expression	Ans	#6 → #54
Arc length	arc	#24
Average value	avg	#23
Closing bracket). Closing brackets at the end of a line may be omitted	)	#31
Comma / argument separator (between function arguments, list items or coordinates)	,	#32
Cosine (current angle unit)	cos	#28
Cursor ▲ — up / previous history line / previous item	▲	#8
Cursor ▶ — move right in the line	▶	#11
Cursor ▼ — down / next history line / next item	▼	#9
Cursor ◀ — move left in the line	◀	#10
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#53
Definite integral of f	∫	#18
DEL — delete the character at the cursor	DEL	#40
Derivative of f	d/dx	#17
Digit 0	0	#52
Digit 1	1	#47
Digit 2	2	#48
Digit 3	3	#49
Digit 4	4	#42

Digit 5	5	#43
Digit 6	6	#44
Digit 7	7	#37
Digit 8	8	#38
Digit 9	9	#39
Division ÷	÷	#46
Evaluate f	F(	#25
e^x — exponential	ex	#29
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	=	#56
GRAPH — draw the graph	PLOT	#5
HOME — return to the main calculation screen	HOME	#12
Left Riemann sum	L∑	#19
Midpoint sum	M∑	#20
Multiplication ×	×	#45
Natural logarithm ln	ln	#6 → #29
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#54
Newton root from x_0	Newt	#21
ON — power on	ON	#13
Opening bracket (	(	#30
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#26
Riemann, Newton, Taylor, tangent	CALC	#7
Right Riemann sum	R∑	#6 → #19
Second derivative	d^2	#6 → #17
sel	SEL	#15
SETUP — angle unit, number format, display options	MODE	#16
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#6

Sine (uses the current angle unit)	sin	#27
Square root √	√	#6 → #26
Square x^2	x^2	#34
Subtraction −	−	#51
TABLE — table of values	TABLE	#6 → #5
Tangent (current angle unit)	tan	#36
Taylor polynomial value	Tay	#22
Toggle derivative graph	f′ ON	#6 → #1
TRACE — move along a graph	TRACE	#4
Trapezoid rule	T∑	#6 → #20
Variable x	X	#14
WINDOW — set the graph view	WIN	#2
Y= — enter functions to graph	f(x)	#1
ZOOM — zoom presets	ZOOM	#3
π (3.14159265…) as an exact constant	π	#33

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Calculus visualiser on a monochrome screen. Line styles distinguish f (solid), g (short dash), f′ (long dash) and f″ (dotted).

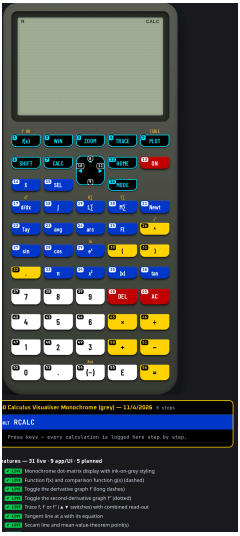
Higher Taylor derivatives use central finite differences, so degrees above 4 are approximate.

Planned (not yet working)

- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 f(x) ⇧ f′ ON	#2 WIN	#3 ZOOM
#4 TRACE	#5 PLOT ⇧ TABLE	#6 SHIFT
#7 CALC	#8 ▲	#9 ▼
#10 ◀	#11 ▶	#12 HOME
#13 ON	#14 X	#15 SEL
#16 MODE	#17 d/dx ⇧ d^2	#18 ∫
#19 L∑ ⇧ R∑	#20 M∑ ⇧ T∑	#21 Newt
#22 Tay	#23 avg	#24 arc
#25 F(	#26 ^ ⇧ √	#27 sin
#28 cos	#29 ex ⇧ ln	#30 (
#31)	#32 ,	#33 π
#34 x^2	#35 |x|	#36 tan
#37 7	#38 8	#39 9
#40 DEL	#41 AC	#42 4
#43 5	#44 6	#45 ×
#46 ÷	#47 1	#48 2
#49 3	#50 +	#51 –

#52 0	#53 .	#54 (−) ⇧ Ans
#55 E	#56 =	

U-45 CALCULATORS · Calculator #40 · cut out and keep